

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following do you agree with?

1 / 1 point

- ☐ Face verification requires K comparisons of a person's face.
- ☒ Face recognition requires K comparisons of a person's face.
- ☐ Face recognition requires comparing pictures against one person's face.

✓ Nice work

Correct, in face recognition we compare the face of one person to K to classify the face as one of those K or not.

2. Why is the face verification problem considered a one-shot learning problem? Choose the best answer.

1 / 1 point

- ☐ Because we have only have to forward pass the image one time through our neural network for verification.
- ☐ Because we are trying to compare to one specific person only.
- ☒ Because we might have only one example of the person we want to verify.
- ☐ Because of the sensitive nature of the problem, we won't have a chance to correct it if the network makes a mistake.

✓ Nice work

Correct. One-shot learning refers to the amount of data we have to solve a task.

3. In order to train the parameters of a face recognition system, it would be reasonable to use a training set comprising 100,000 pictures of 100,000 different persons.

1 / 1 point

3. In order to train the parameters of a face recognition system, it would be reasonable to use a training set comprising 100,000 pictures of 100,000 different persons.

1 / 1 point

☒ False☐ True **Nice work**

Correct, to train a network using the triplet loss you need several pictures of the same person.

4. In the triplet loss:

1 / 1 point

$$\max \left(\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha, 0 \right)$$

Which of the following are true about the triplet loss? Choose all that apply.

☐ α is a trainable parameter of the Siamese network.☒ $f(A)$ represents the encoding of the Anchor. **Nice work**

Correct. f represents the network that is in charge of creating the encoding of the images, and A represents the anchor image.

☒ We want that $\|f(A) - f(P)\|^2 < \|f(A) - f(N)\|^2$ so the negative images are further away from the anchor than the positive images. **Nice work**

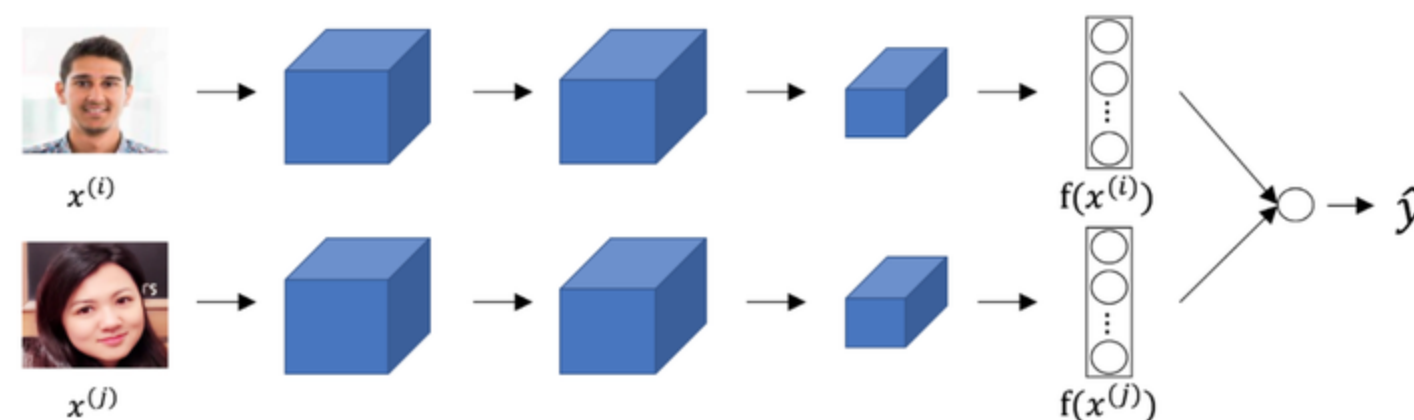
Correct. Being a positive image the encoding of P should be close to the encoding of A .

☐ A the anchor image is a hyperparameter of the Siamese network.

☐ A the anchor image is a hyperparameter of the Siamese network.

5. Consider the following Siamese network architecture:

1 / 1 point



Which of the following do you agree with the most?

- ☐ This depicts two *different* neural networks with different architectures, although we use the same drawing.
- ☐ Although we depict two neural networks and two images, the two images are combined in a single volume and pass through a single neural network.
- ☐ The two neural networks depicted in the image have the same architecture, but they might have different parameters.
- ☒ The upper and lower neural networks depicted have exactly the same parameters, but the outputs are computed independently for each image.

✓ Nice work

Correct. Both neural networks share the same weights, and each image passes through the neural network in an independent manner.

6. You train a ConvNet on a dataset with cats, dogs, birds, and other types of animals. You try to find a filter that strongly responds to horizontal edges. You are more likely to find this filter in layer 6 of the network than in layer 1. True/False?

1 / 1 point

☒ False☐ True **Nice work**

Correct. Edges are a very low-level feature, thus it is more likely to find such a feature detector in the first layers of the network.

7. Neural style transfer is trained as a supervised learning task in which the goal is to input two images (x), and train a network to output a new, synthesized image (y).

1 / 1 point

☐ True☒ False **Nice work**

Yes, Neural style transfer is about training the pixels of an image to make it look artistic, it is not learning any parameters.

8. In neural style transfer the content loss J_{cont} is computed as:

1 / 1 point

$$J_{cont}(G, C) = \|a^{[l](C)} - a^{[l](G)}\|^2$$

Where $a^{[l](k)}$ is the activation of the l -th layer of a ConvNet trained for classification. We choose l to be a very high value to use compared to the more abstract activation of each image. True/False?

☒ False☐ True

learning any parameters.

8. In neural style transfer the content loss J_{cont} is computed as:

1 / 1 point

$$J_{cont}(G, C) = \|a^{[l](C)} - a^{[l](G)}\|^2$$

Where $a^{[l](k)}$ is the activation of the l -th layer of a ConvNet trained for classification. We choose l to be a very high value to use compared to the more abstract activation of each image. True/False?

- ☒ False
- ☐ True

✔ Nice work

Correct. We don't use a very deep layer since this will only compare if the two images belong to the same category.

9. In neural style transfer, what is updated in each iteration of the optimization algorithm?

1 / 1 point

- ☐ The neural network parameters
- ☐ The pixel values of the content image C
- ☐ The regularization parameters
- ☒ The pixel values of the generated image G

✔ Nice work

Yes, neural style transfer is different from many of the algorithms you've seen up to now, because it doesn't learn any parameters; instead it learns directly the pixels of an image.

10. You are working with 3D data. The input "image" has size $64 \times 64 \times 64 \times 3$, if you apply a convolutional layer with 16 filters of size $4 \times 4 \times 4$ zero padding and stride 2. What is the size of the output volume?

1 / 1 point

Correct. We don't use a very deep layer since this will only compare if the two images belong to the same category.

9. In neural style transfer, what is updated in each iteration of the optimization algorithm?

1 / 1 point

- ☐ The neural network parameters
- ☐ The pixel values of the content image C
- ☐ The regularization parameters
- ☒ The pixel values of the generated image G

✔ Nice work

Yes, neural style transfer is different from many of the algorithms you've seen up to now, because it doesn't learn any parameters; instead it learns directly the pixels of an image.

10. You are working with 3D data. The input "image" has size $64 \times 64 \times 64 \times 3$, if you apply a convolutional layer with 16 filters of size $4 \times 4 \times 4$, zero padding and stride 2. What is the size of the output volume?

1 / 1 point

- ☐ $61 \times 61 \times 61 \times 14$.
- ☐ $64 \times 64 \times 64 \times 3$.
- ☐ $31 \times 31 \times 31 \times 3$.
- ☒ $31 \times 31 \times 31 \times 16$.

✔ Nice work

Correct, we can use the formula $\lfloor \frac{n^{[l-1]} - f + 2 \times p}{s} \rfloor + 1 = n^{[l]}$ to the three first dimensions.